Smart Parking Platform

Project Kickoff Document

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Vision and Scope

Project Acquisition

The acquisition for this project happened on the basis of a client demand-driven approach rather than on a market-driven basis. A regional consortium comprising a municipality transportation department and some private parking facilities owners issued a Request for Proposals (RFP) for one platform for modernizing drivers' search for parking spots in garages, booking and payment process for those spots, and monitoring and pricing their facility by operators.

$oftware Corp's development team presented a proposal for this RFP based on its past experience in developing for mobility services sector. The proposal started with analysis of pain points of the consortium – constant downtown traffic due to circling of drivers looking for parking spots, and lack of real-time data on facility occupancy for operators – and not with a traditional sales presentation. The contract was awarded after a competitive tender and discussion of the scope.

Acquisition summary: Client-driven, through RFP / competitive tender. Revenue is linked to milestone-based delivery with a defined scope and timeframe, i.e. it is project-based (not product).

Company Introduction

$oftware Corp is a hypothetical software development agency working with theoretically unlimited amounts of engineering capacity and funding. The company focuses on developing end-to-end, cloud-based platforms to connect common customers with the owners of physical infrastructure – from a disparate array of offline activities to one cohesive set of digital products.

Not being a regular software agency, $oftware Corp follows an outlined go-to-market strategy where the company specializes in a few verticals, proves its competence publicly and wins customers via an inbound research-led approach. One of such verticals is urban mobility and smart-city infrastructure. Thus, a Smart Parking Platform is a natural product for the studio.

Project Overview

The Smart Parking Platform is an application that will enable users to find, reserve and pay for parking spots in parking lots that are connected to the service. It is delivered using both the web portal and the mobile app, and allows parking availability, navigation and management services to both parking spots' users and parking lot operators.
The engagement is a project: to develop a plan and manage the creation of the system mentioned above – gathering of requirements, working with stakeholders, planning, budgeting, risk and team management. The deliverable is the developed product for a particular customer.

Business Need

People who need parking spots waste a lot of time looking for them, which results in additional traffic and wasted time and productivity. At the same time, parking lot operators do not have enough instruments to control their lots' occupancy rates and pricing.

Vision Statement

The Smart Parking Platform is an online parking solution for drivers and parking operators who lose valuable time and money through inefficient, offline parking. This system brings together real-time parking space availability, guaranteed parking reservations, payments, and operators' dashboard

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Project Scope

In Scope

• The ability to register users and authenticate them as drivers, operators, or administrators.
• The real-time parking-space availability of the participating parking facilities.
• The interactive map showing available parking facilities.
• Parking reservation functionality with confirmation.
• Digital payments via third-party payment providers.
• History of reservation and receipt of the same.
• The operators' dashboard of parking facilities (facilities management, pricing, and occupancy).
• Occupancy analytics.
• Notification and alert functionality for the users (alerts regarding expiration, available parking spaces).
• Integration with the external map / navigation services.

Out of Scope

• The manufacture and installation of parking facilities' hardware (gates, sensors, LPR cameras); the platform uses it but doesn't make it.
• On-street metered parking regulation and citations issuing.
• Monthly parking lease contract management.
• Payment processing and payment institution activity (handled by third party).
• Valet and autonomous vehicle management.

Stakeholders

• Driver/passenger and car owners – main end users who do searches, reservations, and payments.
• Parking facilities owners/operators – responsible for management of parking facilities, fees, and availability of parking spaces.
• Traffic departments – focused on congestion control and traffic rules.
• Platform administrators – responsible for user management, permissions, and system configuration.
• Finance department employees – those who will be doing reconciliation of payments and revenue reports analysis.
• Mobile application users – subset of drivers who use mobile application.
• Payment processing partners – outside vendors providing transaction processing functionality.

Development Teams

The engagement involves the following task-oriented cross-functional teams:
Mobile application team
Web application team

Backend / API team
Mapping and location services team
Payment integration team
Quality assurance and testing team
Roles are strongly tied to the scope and timelines and will be re-assigned post-delivery.

Project Constraints

• Unlimited theoretical research funding (not financially based but quality- and scope-related).
• The same limited timeframe throughout the semester with milestones.
• Making use of the services provided by third parties (payment, maps, sensors' feeds).
• Conforming to security and privacy standards, city ordinances and relevant legislation.

4. Research: Existing Software Landscape

Prior to developing the specific details of the platform, we conducted a competitive analysis of the market for existing parking software to determine strengths, weaknesses, and areas where there is room for differentiation in the offering.

4.1 Driver-Facing Consumer Apps

This type of application is used by drivers to find, reserve and pay for parking spaces. Major players in the market include ParkMobile, SpotHero, ParkWhiz, PayByPhone, and BestParking. ParkMobile focuses on on-street metered parking and municipal parking lot payments and integrates with the city systems of hundreds of cities in the USA. SpotHero and ParkWhiz provide for pre-reservation of off-street parking spaces in private garages and lots usually with a price much lower than the drive-in price. PayByPhone provides pay-per-use meter services and serves as the official parking app in many cities.

Weakness common to all of these apps is that their inventory is made up almost exclusively of commercial garages and lots, and a common issue in the reviews is reservation being ignored by the parking attendant who either does not know about the reservation or reports that the garage is already full.

4.2 Operator-Facing Management Platforms

In the supply chain, management platforms like AirGarage, FLASH, T2 Systems, Parking BOXX, and SkiData offer operators a cloud-based control panel to manage a parking facility: set prices, control access credentials, handle transactions and report on revenue and occupancy. Contemporary solutions in this segment offer license plate recognition (LPR), real-time occupancy and monitoring, enforcement, and dynamic pricing that adjusts the price based on occupancy levels.

Weakness of these platforms is that they are designed to serve operators and property owners rather than drivers and therefore do not offer drivers any unified view of occupancy and available spaces.

4.3 Competitive Comparison

4.4 Shortcomings and Opportunities

• There is no platform that brings together live driver-facing availability with the true occupancy data of the operator, which is at the heart of the "reservation not honored" issue.
• Driver apps and operator platforms are two different products, hence, the prices, availability, and bookings aren’t always in sync.
• Each operator optimizes independently for their lot; there is no neutral, cross-operator marketplace that offers live spots for drivers.

4.5 What Sets Our Platform Apart

• Two-sided in nature: One platform catering to the needs of both drivers and operators, using a single source of the truth for space availability.
• Real-time, space-based availability linked to the actual occupancy data of the operator, thereby ensuring a live spot for every booking made.
• One-stop shop for reservations, payments, invoices, and navigation for the driver as well as occupancy reports and pricing for the operator.
• Cross operator neutral marketplace: Drivers compare and book from multiple operators; operators get demand that wouldn't be possible otherwise.

Software Requirements Specification (SRS)

5.1 Introduction

5.1.1 Purpose

The purpose of this document is to define the software requirements for the Smart Parking Platform: web and mobile application allowing users to find, reserve, and pay for parking at garage parking in real time as well as to monitor and manage parking lots for the operators.

5.1.2 Scope

The product encompasses: registration and authentication, real-time availability, a map, reservation system, online payments, transactions history, operator panel, occupancy statistics, notifications, and integration with mapping and payments services. Manufacturing of hardware devices, enforcement of regulations on-street parking, and providing payments service are explicitly out of scope (see section 3.4).

3.2.1 Users

• Drivers / mobile application users - search, book, pay, navigate and manage vehicles and receipts.
• Parking facility managers - manage parking facilities and slots, define pricing policies and monitor occupancy rates.
• Finance and billing department - handle payments, reconciliations and generation of income reports.
• System Administrators - manage users, roles and configurations of the system.
• City transportation departments - consume aggregated/occupancy reports according to agreement.

3.2.2 System Environment

System is offered as native mobile applications (iOS and Android) and web applications that operate using cloud-hosted services and APIs. System incorporates services from third parties for navigation, payment processing and real time feeds from IoT occupancy sensors and camera/LPR devices.

3.2.3 Constraints

• The solution should incorporate services from third parties and depend on their availability.
• The solution should be compliant with PCI-DSS and all other privacy related legal requirements.
• The solution should be designed and implemented within semester-based timeframe.

3.2.4 Assumptions & Dependencies

• Users and devices are connected to Internet or Intranet.
• Park facilities have occupancy sensors capabilities or capacity information.
• Third party providers of payment, mapping and sensors have APIs available and defined.

Functional Requirements

ID	Requirement
FR1	The system shall permit user registration of an account with the system using either their email or single sign-on services that are compatible.
FR2	The system shall provide secure log-in authentication, including password reset and multi-factor authentication.
FR3	The system shall display the live availability of parking spots at the selected location during the specified period of time.
FR4	The system shall show the available parking locations on an interactive map.

FR5	The system shall allow drivers to perform search and filters of parking places based on location, prices, distance and type (EV parking, etc.).
FR6	The system shall allow drivers to reserve a parking spot for the selected date and time.
FR7	The system shall process electronic payments for reservation and session via third-party integrated payment processors.
FR8	The system shall provide an electronic ticket and receipt for every successfully made reservation.
FR9	The system shall maintain and display user reservation and receipt history.
FR10	The system shall allow drivers to extend or cancel their reservation in accordance with the operators' policies.
FR11	The system shall provide notifications and alerts (reservations confirmation, reminder and expiration, price or availability change).
FR12	The system shall provide navigation to the reserved parking space through integrated mapping/navigation providers.
FR13	The system shall allow drivers to store multiple vehicle profiles (license plates, car make and model).
FR14	The system shall provide a comprehensive administrative interface for parking operators to handle management operations related to parking facility.
FR15	The system shall have the ability to apply dynamic pricing policies configurable by operators.
FR16	The system shall be able to generate occupancy and utilization reports for operators and finance department.
FR17	The system shall ingest occupancy information in real-time from IoT sensors and license plate detection cameras.
FR18	The system shall permit system administrators to administer users, roles and access permissions.

Non-functional Requirements

ID	Requirement
NFR1	The system is supposed to accommodate up to 10,000 simultaneous users at launch and scale horizontally depending on load.
NFR2	Changes in space availability are reflected in the system within 2 seconds of sensor data reception.
NFR3	The platform has to guarantee 99.9% uptime as calculated on a monthly basis.

NFR4	Map and availability screen are expected to load within 3 seconds via 4G network on a common phone.
NFR5	Payment transactions must take place via encrypted communication channel.
NFR6	All payment and personally identifiable information must be protected via encryption during both transmission (TLS 1.2+) and rest (AES-256).
NFR7	The system must follow all the required legislation concerning data privacy and local city laws.
NFR8	Mobile and desktop versions of the applications have to meet WCAG 2.1 AA guidelines regarding their accessibility.
NFR9	The apps should support current and previous iOS/Android major release as well as all evergreen browsers.
NFR10	The security events are to be logged and audit logs stored for not less than one year.

Use Cases

UC-01: Register Account
Actors: Prospective Driver
Preconditions: The user has downloaded the mobile application or opened the web application but has not yet created an account.
Main Flow:
1. The user chooses "Create Account."
2. The user provides email, password, and profile information, or use single sign-on.
3. The system performs validation and checks for duplicate account.
4. The system creates the user account and sends verification information.
5. The user verifies the user account.
Postconditions: The user account exists, is verified and the user logged in.
Traceability: FR1, FR2
UC-02: Log In
Actors: Registered Driver
Preconditions: The user exists and already verified the account but is logged out.
Main Flow:
1. The user provides credentials or use single sign-on.
2. The system authenticates the user with multi-factor authentication if enabled.
3. The system initiates secure session.
Postconditions: The user authenticated and reaches the home screen.
Traceability: FR2
UC-03: Search Available Parking Options
Actors: Driver
Preconditions: The user logged in and provided location access.
Main Flow:

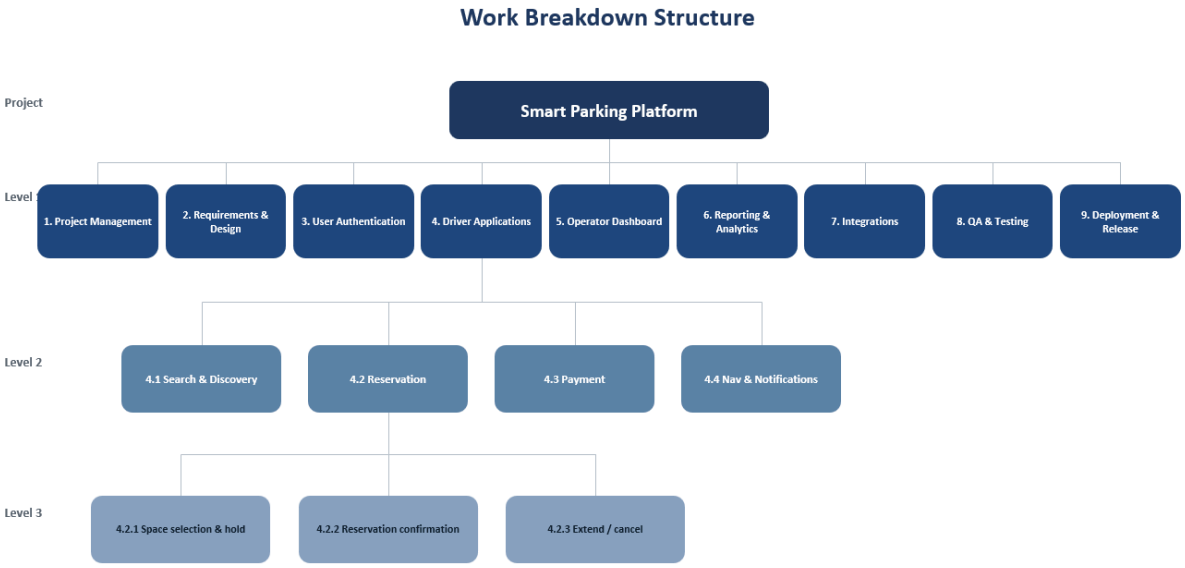
1. The user inputs a destination or uses the current location.
2. The user enters the date and expected stay time.
3. The system searches for real-time available options and prices.
4. The system provides a ranked list of nearby available parking options.
Postconditions: Available parking options list is provided.
Traceability: FR3, FR5
UC-04: View Available Spaces on Interactive Map
Actors: Driver
Preconditions: The user has done the search or opened the map view.
Main Flow:
1. The system shows nearby available parking spots on the interactive map.
2. The system paints each spot with different color according to live availability.
3. The user navigates (pans, zooms) the map and selects any available spot for more information.
Postconditions: The system displays live parking availability.
Traceability: FR3, FR4
UC-05: Book a Space
Actors: Driver
Preconditions: The user picked one available spot and is logged in.
Main Flow:
1. The user picks the parking space and confirms date, time and vehicle type.
2. The system puts on hold the space until the booking is canceled.
3. The system estimates the overall price and presents them to the user.
4. The user confirms the booking.
Postconditions: Parking spot booked and reserved for user specific time window.
Traceability: FR6
UC-06: Pay for a Reservation
Actors: Driver
Preconditions: User has some pending reservation(s), open session and the payment option (saved before).
Main Flow:
1. User chooses payment method.
2. The system forwards the transaction to the third-party payment processor.
3. The payment processor authorizes the payment and send confirmation back.
4. The system stores the record of the payment.
Postconditions: Payment process is finished and the reservation is confirmed.
Traceability: FR7, NFR5
UC-07: Provide Digital Pass & Receipt
Actors: Driver
Preconditions: Parking reservation is paid.
Main Flow:
1. The system generates the parking digital pass which includes entry time details.
2. The system generates the itemized receipt.
3. The system delivers both documents to the user via app and by email.

Postconditions: The user receives a valid parking pass and receipt.
Traceability: FR8
UC-08: View My Reservations List
Actors: Driver
Preconditions: User logged in and has made at least one reservation before.
Main Flow:
1. User opens "My Reservations" tab.
2. The system generates list with user past and upcoming reservations including receipt document.
3. User selects any past reservations to download its receipt.
Postconditions: The user can see the past and upcoming reservations and download receipts.
Traceability: FR9
UC-09: Extend Active Session
Actors: Driver
Preconditions: The user has the active parking session.
Main Flow:
1. The user opens "Extend" section.
2. The system checks the availability of the space after the current one.
3. The system calculates the additional fee and asks user confirmation.
4. User accepts the additional fee and pays it, then the system extends the parking session.
Postconditions: The parking session is extended.
Traceability: FR10, FR7
UC-10: Cancel a Reservation
Actors: Driver
Preconditions: The user have any upcoming reservation which could be cancelled.
Main Flow:
1. User chooses to "Cancel Reservation."
2. The system uses operator’s cancellation policy and refund rules.
3. User confirms cancellation.
4. The system cancels the parking spot reservation and release it back in available status.
Postconditions: The reservation is canceled.
Traceability: FR10
UC-11: Get Notification & Alerts
Actors: Driver
Preconditions: User logged in and have notification setting enabled.
Main Flow:
1. The system detects some event which need to notify the user about it (confirmation, reservation expiration or other).
2. The system builds up the notification content.
3. The system delivers this notification to user via push-notification, email or SMS as he/she requested.
Postconditions: The user is notified.
Traceability: FR11
UC-12: Find Directions to Parking Spot

Actors: Driver
Preconditions: User have some reservation confirmed.
Main Flow:
1. User chooses "Directions" button.
2. The system sends the destination data to the integrated maps service (Google or another).
3. The system generates and shows step-by-step route.
Postconditions: The user will have directions to the parked vehicle.
Traceability: FR12
UC-13: Manage Vehicles Information
Actors: Driver
Preconditions: User logged in.
Main Flow:
1. User goes to "My Vehicles" tab.
2. User changes vehicle information (plate number, car make, etc).
3. The system validates the information and stores.
Postconditions: The user manages vehicles list.
Traceability: FR13
UC-14: Occupancy Dashboard Monitoring
Actors: Parking Operators
Preconditions: The user has logging access as an operator and facility permissions to see facilities occupancy information.
Main Flow:
1. The user opens "Occupancy Dashboard."
2. The system shows facilities current occupancy level from sensor data.
3. The user drills down for particular facility for further details.
Postconditions: The operator will know the current occupancy in his facilities.
Traceability: FR14, FR17, NFR2
UC-15: Configure Dynamic Pricing
Actors: Parking Operators
Preconditions: The user has access to the facility configuration as an operator and can edit pricing data.
Main Flow:
1. The user opens the facility pricing section.
2. The user defines base rate and dynamic pricing rules depending on the demands.
3. The system validates rules and generates price preview.
4. The user publishes new pricing.
Postconditions: New pricing rules are published.
Traceability: FR15
UC-16: Generate Report
Actors: Finance/User
Preconditions: The user logged in and has permission to generate reports.
Main Flow:
1. User selects report type, facilities for and time range for report generation.
2. The system aggregates occupancy, utilization and revenues information.

3. The system renders report and provides export (PDF, CSV files).
Postconditions: Report generated and exported.
Traceability: FR16
UC-17: Add / Edit Parking Facilities
Actors: Parking Operator
Preconditions: User has logging access and administrative access to the operator section and add facilities.
Main Flow:
1. The user selects "Add Facility."
2. The user inputs location, number of spots and zoning scheme of the facility, spot types and other.
3. The user connects sensor devices with parking spots.
4. The system validates and adds the facility into database.
Postconditions: The new facility is added and accessible by all drivers.
Traceability: FR14, FR17
UC-18: Managing Users & Roles
Actors: System Administrator
Preconditions: User logged in with higher privileges.
Main Flow:
1. The administrator opens users management tab.
2. The administrator manages (add, delete, modify) user accounts and assign different roles.
3. The system applies user roles with respective permissions and logs this operation.
Postconditions: User accounts and their roles are managed and permissions are applied.
Traceability: FR18, NFR10

Work Breakdown Structure (WBS)



Complete WBS Outline

1 Project Management

- **1.1 Planning**
  - **1.1.1** Project charter & kickoff
  - **1.1.2** Vision & scope document
  - **1.1.3** WBS & schedule
- **1.2 Monitoring & Control**
  - **1.2.1** Status reporting
  - **1.2.2** Risk management

2 Requirements & Design

- **2.1 Requirements**
  - **2.1.1** Requirements elicitation
  - **2.1.2** SRS document
  - **2.1.3** Use-case modeling
- **2.2 Design**
  - **2.2.1** System architecture
  - **2.2.2** UI/UX design
  - **2.2.3** Database design
  - **2.2.4** API design

3 User Authentication & Account Management

- **3.1 Registration**
  - **3.1.1** Email / SSO signup
  - **3.1.2** Account verification
- **3.2 Login & Session**

- **3.2.1** Secure login
- **3.2.2** Multi-factor authentication
- **3.2.3** Session management

3.3 Profile Management

- **3.3.1** User profile
- **3.3.2** Vehicle profiles

4 Driver Applications (Mobile & Web)

4.1 Parking Search & Discovery

- **4.1.1** Location search & filters
- **4.1.2** Interactive map view
- **4.1.3** Real-time availability display

4.2 Reservation

- **4.2.1** Space selection & hold
- **4.2.2** Reservation confirmation
- **4.2.3** Extend/cancel

4.3 Payment

- **4.3.1** Payment method management
- **4.3.2** Payment processing
- **4.3.3** Digital pass & receipt

4.4 Navigation & Notifications

- **4.4.1** Turn-by-turn navigation
- **4.4.2** Alerts & reminders

5 Operator Dashboard & Garage Monitoring

5.1 Facility Management

- **5.1.1** Add/edit/remove facility
- **5.1.2** Space & zone configuration

5.2 Occupancy Monitoring

- **5.2.1** Live occupancy dashboard
- **5.2.2** Sensor / camera data integration

5.3 Pricing

- **5.3.1** Base-rate configuration
- **5.3.2** Dynamic pricing rules

6 Reporting & Analytics

6.1 Occupancy & Utilization Reports

6.2 Financial Reports

- **6.2.1** Revenue reporting
- **6.2.2** Export (PDF / CSV)

6.3 Administration

- **6.3.1** User & role management
- **6.3.2** Audit logging

7 Integrations

7.1 Payment Gateway Integration

7.2 Mapping / Navigation Integration

7.3 IoT Sensor / LPR Integration

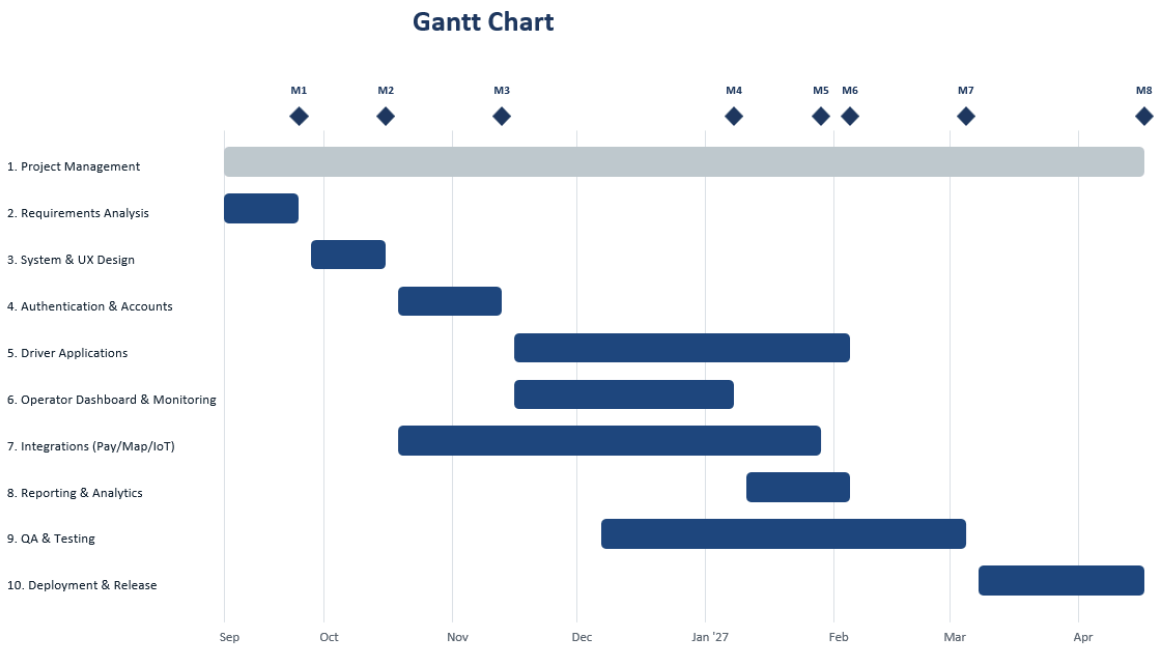
8 Quality Assurance & Testing

8.1 Test Planning
8.2 Functional & Integration Testing
8.3 Security & Performance Testing
8.4 User Acceptance Testing
9 Deployment & Release
9.1 Environment Setup
9.2 Beta Release
9.3 Production Release
9.4 Handoff & Documentation

Project Schedule & Timeline

Task / Phase	Start	End	Duration	Pred.	Dep.
1. Project Management	Sep 7, 2026	Apr 17, 2027	32 wks		Spans project
2. Requirements Analysis	Sep 7, 2026	Sep 25, 2026	3 wks	Kickoff	FS
3. System & UX Design	Sep 28, 2026	Oct 16, 2026	3 wks	2	FS
4. Authentication & Accounts	Oct 19, 2026	Nov 13, 2026	4 wks	3	FS
5. Driver Applications	Nov 16, 2026	Feb 5, 2027	12 wks	4	FS
6. Operator Dashboard & Monitoring	Nov 16, 2026	Jan 8, 2027	8 wks	4	SS
7. Integrations	Oct 19, 2026	Jan 29, 2027	15 wks	3	FS
8. Reporting & Analytics	Jan 11, 2027	Feb 5, 2027	4 wks	6	FS
9. QA & Testing	Dec 7, 2026	Mar 5, 2027	13 wks	5	SS
10. Deployment & Release	Mar 8, 2027	Apr 17, 2027	6 wks	9	FS

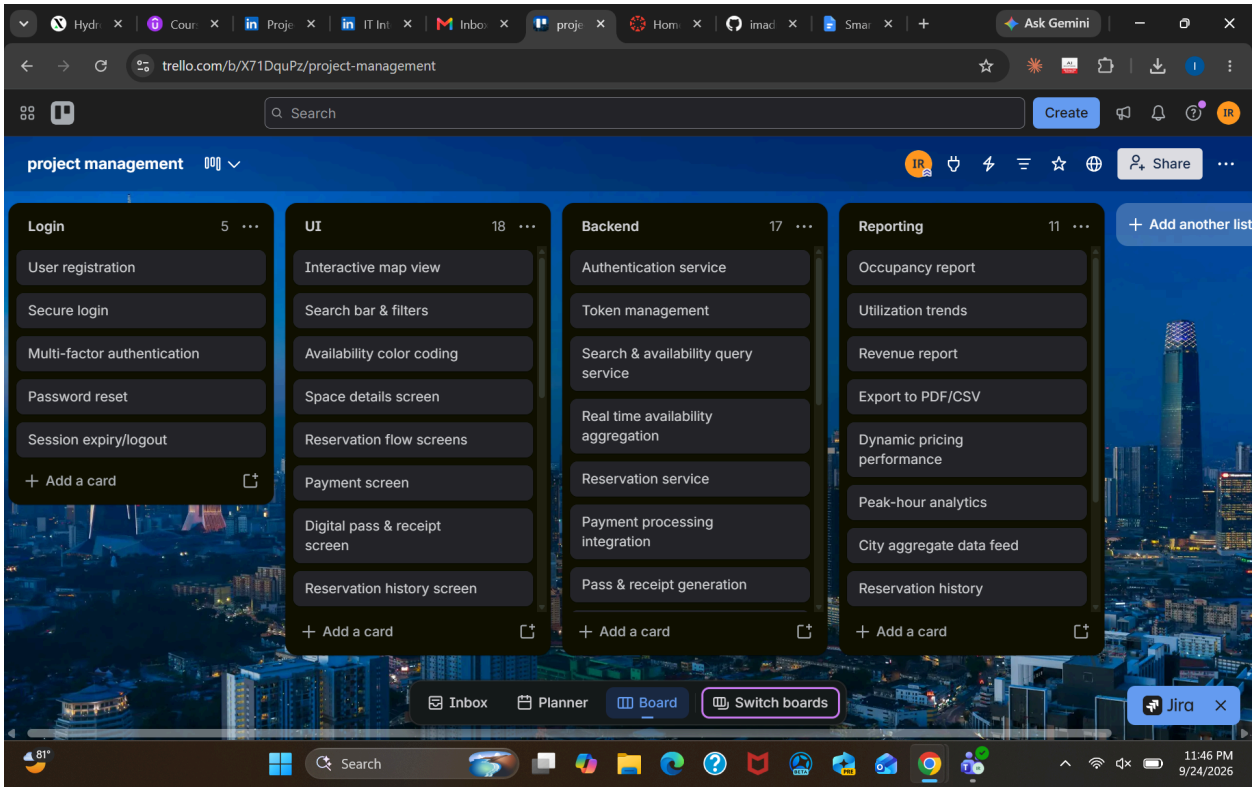
GANTT Chart



Milestones

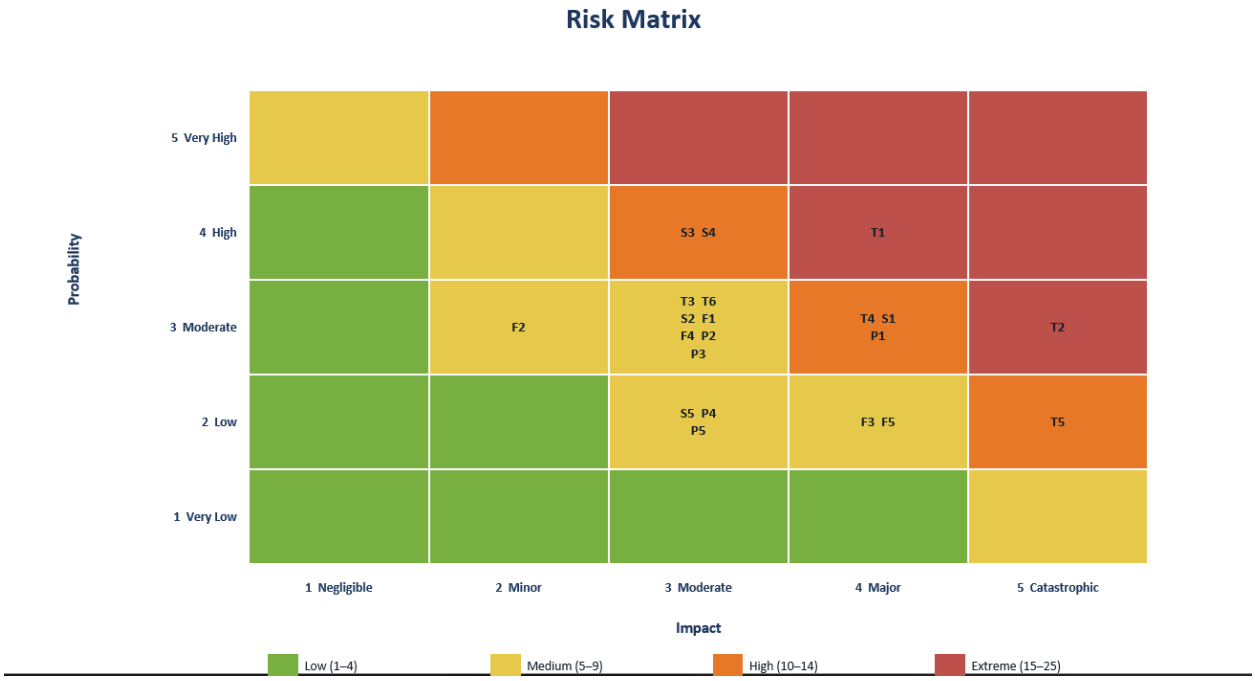
ID	Milestone	Target Date
M1	Requirements baseline	Sep 25, 2026
M2	Design complete	Oct 16, 2026
M3	Authentication module complete	Nov 13, 2026
M4	Operator dashboard complete	Jan 8, 2027
M5	Integrations complete	Jan 29, 2027
M6	Driver app MVP	Feb 5, 2027
M7	Beta release	Mar 5, 2027
M8	Production release/handoff	Apr 17, 2027

Trello Board



Link for full: https://trello.com/b/X71DquPz/project-management

Risk Matrix



Risk register

ID	Risk Description	P	I	Score	Rating	Owner	Response	Status / Notes
Technical Risks								
T1	Real-time IoT/sensor data feed is inaccurate, so the app incorrectly displays availability.	4	4	16	**Extreme**	Mapping & Location Lead	Mitigate	Open – predictive fall-back mechanism plus sensor health checks.
T2	Changes made to the third-party payment gateway API, or an outage occurs, disrupting payments.	3	5	15	**Extreme**	Payment Integration Lead	Transfer / Mitigate	Open – provider adapter + backup provider.
T3	Rate limit applied to or deprecation of mapping/navigating API.	3	3	9	**Medium**	Mapping & Location Lead	Mitigate	Open – caching results; provider-agnostic interface.

T4	The system crashes while under peak concurrent load (10,000+ users).	3	4	12	**High**	Backend Lead	Mitigate	Open – test load early; auto-scaling; cache hot path.
T5	Security issue results in exposure of personal and payment data.	2	5	10	**High**	Security / Backend Lead	Avoid / Mitigate	Open – PCI-DSS, encryption, pen-testing, audits.
T6	Camera/LPR accuracy affected by lighting/weather conditions.	3	3	9	**Medium**	Mapping & Location Lead	Accept / Mitigate	Open – confidence threshold; fall back to sensors.
Schedule Risks								
S1	Design phase delay causes further delays to all development phases.	3	4	12	**High**	Project Manager	Mitigate	Open – buffer plus parallel approach.
S2	Access/credentials for third-party APIs is not available on time.	3	3	9	**Medium**	Project Manager	Mitigate	Open – request access in Sprint 1.
S3	Integration and testing phases were underestimated in the original timeline.	4	3	12	**High**	QA Lead	Mitigate	Open – explicit integration/quality assurance phases.
S4	Requests by stakeholders change the scope in the middle of the sprint.	4	3	12	**High**	Product Owner	Avoid	Open – change management; guard sprint commitment.
S5	Delivery delay of IoT pilot sensor hardware.	2	3	6	**Medium**	Project Manager	Mitigate	Open – order early; simulate feeds in parallel.
Financial Risks								
F1	Time or labor-related budget overrun.	3	3	9	**Medium**	Project Manager	Mitigate	Open – track burn vs. budget; re-estimate every sprint.
F2	Increased costs of third-party services (payment fees, map API) than expected.	3	2	6	**Medium**	Project Manager	Mitigate	Open – cost model; billing triggers.

F3	Delayed payment by the client of the milestone payment.	2	4	8	**Medium**	Project Manager	Transfer	Open – milestone agreements; bill early.
F4	Increased sensor hardware/device integration costs than expected.	3	3	9	**Medium**	Backend Lead	Mitigate	Open – pilot small; bulk purchasing.
F5	Legal and financial penalties related to compliance/privacy issues.	2	4	8	**Medium**	Security Lead	Avoid	Open – legal assessment; build according to ordinance and privacy laws.
People Risks								
P1	Departure of a key member of the development team.	3	4	12	**High**	Project Manager	Mitigate	Open – cross-training; documentation; no single point of failure.
P2	Lack of expertise in IoT/real-time systems.	3	3	9	**Medium**	Project Manager	Mitigate	Open – training; targeted hire/consultant.
P3	Communication/collaboration problems between teams.	3	3	9	**Medium**	Scrum Master	Mitigate	Open – scrum-of-scrums; joint boards.
P4	Burnout of the team because of tight deadline.	2	3	6	**Medium**	Scrum Master	Mitigate	Open – sustainable pace; watch load.
P5	Availability of stakeholders.	2	3	6	**Medium**	Project Manager	Mitigate	Open – fixed cadence reviews; clear decision-makers.

Communication Plan

Meeting Cadence

Meeting	Audience / Team	Frequency	Purpose
Daily stand-up	Each delivery team (Mobile, Web, Backend, Mapping, Payment, QA)	Daily (15 min)	Sync on yesterday/today/blockers.
Backlog refinement	Product Owner, team leads	Weekly	Groom and estimate upcoming stories.
Sprint planning	Delivery team, Product Owner	Each sprint (biweekly)	Commit the sprint backlog.
Scrum-of-scrums	Team leads, Scrum Master	2× per week	Resolve cross-team dependencies.
Sprint review/demo	Teams, stakeholders	End of sprint (biweekly)	Demo the working increment.
Sprint retrospective	Each team	End of sprint	Improve process (start/stop/continue).
Risk review	PM, Scrum Master, leads	Biweekly	Review and update the risk register.
Steering/stakeholder meeting	PM, client, city dept.	Monthly	Big picture: scope, budget, risks, decisions.

Reporting Methods

Report	Audience	Frequency	Channel / Content
Status of board / tasks	Everyone	Throughout	Trello board - live status of cards.
Burndown chart & velocity	Team → PM	Every sprint	Trello dashboard: performance vs. commitments.
Weekly status update	PM → executives	Weekly	Email and Trello dashboard: milestones, risks, budget RAG.
Executive summary	PM → steering committee	Monthly	Presentation deck and demonstration: high-level data and decisions.
Risk register	PM (everyone can see)	Biweekly review	Shared sheet: open risks, ratings, responsible parties, status.

Release notes / changelog	Development team → everyone	Every release	Repo/Wiki - what was released and known bugs.
Incident / bug report	QA / Development	As needed (Kanban)	Ticket and notification: severity, impact, and resolution.

Roles and Resources Plan

Role	Name	Skill	FTE %	Hours	Rate	Cost
Product Owner	John Joe	Senior	40%	512	$90	$46,080
Project Manager	Marcus Lee	Senior	100%	1,280	$85	$108,800
Scrum Master	Elena Ortiz	Mid	60%	768	$75	$57,600
Software Architect	David Kim	Lead	60%	768	$110	$84,480
Backend Developer (Lead)	Sarah Chen	Senior	100%	1,280	$90	$115,200
Backend Developer	Tom Nguyen	Mid	100%	1,280	$80	$102,400
Backend Developer	Aisha Khan	Mid	100%	1,280	$80	$102,400
Mobile Developer	Jason Park	Mid	100%	1,280	$80	$102,400
Mobile Developer	Mia Ross	Mid	100%	1,280	$80	$102,400
Web Developer	Travis Scott	Mid	100%	1,280	$80	$102,400
Web Developer	Grace Lee	Junior	100%	1,280	$65	$83,200
Mapping/Location Engineer	Jack Martinez	Senior	100%	1,280	$90	$115,200
Payment Integration Engineer	Nina Alvarez	Senior	100%	1,280	$95	$121,600
UX Designer	Leo Martin	Mid	70%	896	$70	$62,720
QA Engineer	Hana Suzuki	Mid	100%	1,280	$65	$83,200
QA Engineer	Carlos Mendez	Junior	100%	1,280	$55	$70,400
DevOps Engineer	Yuki Tanaka	Senior	80%	1,024	$90	$92,160
Security Engineer	Fatima Noor	Senior	50%	640	$100	$64,000
Labor subtotal	18 roles					**$1,616,640**

Cost Plan

Item	Category	Basis	Amount	Notes
Labor	Labor	From resource plan	$1,616,640	18 roles, 8-month window
Cloud hosting	Infrastructure	$6,000/mo × 8	$48,000	Compute, storage, egress, monitoring
Software licenses & tools	Tools	one-off + subs	$25,000	IDEs, Jira, design, CI
Third-party APIs	APIs	usage	$20,000	Payment fees, mapping API
IoT sensor / LPR hardware	Hardware	pilot	$35,000	Pilot facility devices
Security (pen-test, audit)	Security	PCI / compliance	$30,000	Pen-test, audit, PCI scope
Training	Training	team	$10,000	IoT/ eal-time upskilling
Non-labor subtotal		sum of direct non-labor	$168,000	
Overhead / benefits	Overhead	25% of labor	$404,160	Taxes & benefits
Subtotal (before contingency)		labor + non-labor + overhead	$2,188,800	
Contingency	Reserve	15% of subtotal	$328,320	Known-unknowns
GRAND TOTAL (BAC)			$2,517,120	Budget at Completion

Budgeted Cashflow

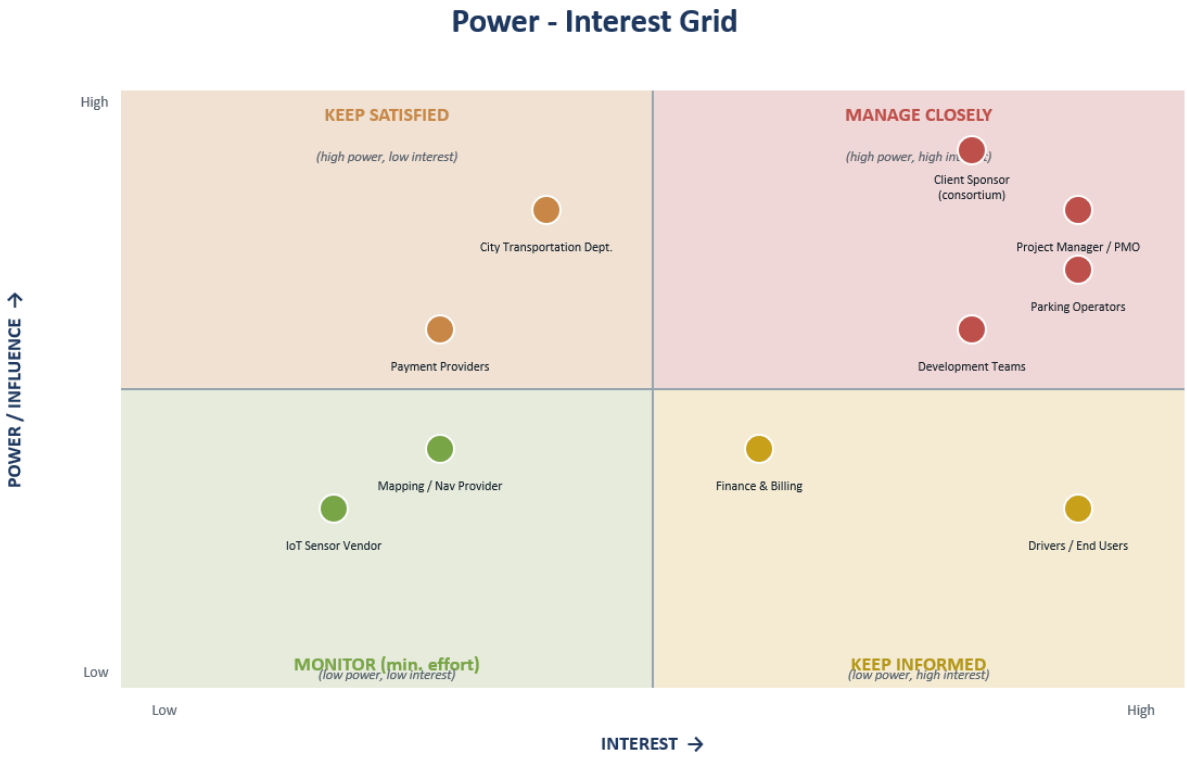
Month	Labor	Non-Labor	Overhead	Contingency	Monthly Total	Cumulative
Sep 2026	$129,331	$27,000	$32,333	$41,040	$229,704	$229,704
Oct 2026	$193,997	$20,000	$48,499	$41,040	$303,536	$533,240
Nov 2026	$226,330	$30,000	$56,582	$41,040	$353,952	$887,192

Dec 2026	$226,330	$25,000	$56,582	$41,040	$348,952	$1,236,144
Jan 2027	$226,330	$20,000	$56,582	$41,040	$343,952	$1,580,096
Feb 2027	$226,330	$10,000	$56,582	$41,040	$333,952	$1,914,048
Mar 2027	$210,163	$30,000	$52,541	$41,040	$333,744	$2,247,792
Apr 2027	$177,829	$6,000	$44,459	$41,040	$269,328	$2,517,120
TOTAL	**$1,616,640**	**$168,000**	**$404,160**	**$328,320**	**$2,517,120**	

RACI Matrix

Roles: PO (Product Owner), PM (Project Manager), Arch (Architect), Dev (development teams), UX, QA, DevOps, and Sec (Security)

Deliverable / Phase	PO	PM	Arch	Dev	UX	QA	DevOps	Sec
Requirements & Analysis	**A**	**R**	**C**	**C**	**C**	**I**	**I**	**C**
System & UX Design	**C**	**A**	**R**	**C**	**R**	**I**	**C**	**C**
Authentication & Accounts	**I**	**A**	**C**	**R**	**I**	**C**	**I**	**C**
Driver Applications	**C**	**A**	**C**	**R**	**C**	**C**	**I**	**I**
Operator Dashboard & Monitoring	**C**	**A**	**C**	**R**	**C**	**C**	**I**	**I**
Integrations (Payment/Mapping/IoT)	**I**	**A**	**C**	**R**	**I**	**C**	**C**	**C**
Reporting & Analytics	**A**	**R**	**C**	**R**	**C**	**C**	**I**	**I**
QA & Testing	**I**	**C**	**C**	**R**	**I**	**A**	**C**	**C**
Deployment & Release	**I**	**A**	**C**	**R**	**I**	**C**	**R**	**C**
Risk & Cost Management	**C**	**A**	**C**	**I**	**I**	**I**	**I**	**C**



Stakeholder Register

Stakeholder	Power	Interest	Quadrant	Engagement Strategy
Client Sponsor (consortium)	High	High	Manage Closely	Supplies funds for project execution and takes ownership of delivery; weekly steering meeting, project management contact
Project Manager / PMO	High	High	Manage Closely	Owns execution; facilitates planning, risk management, and reporting.
Parking Operators	High	High	Manage Closely	Key Paying Customer; part of requirement, demo, and facility setup.
Development Teams	Medium	High	Manage Closely	Build the product; involved through sprint ceremonies and the board.
City Transportation Dept.	High	Low	Keep Satisfied	Regulation/Odinance Authority; get updates from time to time and get signed off on.

Payment Providers	Medium	Low	Keep Satisfied	Key dependency; handle through SLAs and integration points.
Drivers / End Users	Low	High	Keep Informed	Large and very interested audience; communicate through release notes and beta feedback.
Finance & Billing	Medium	Medium	Keep Informed	Manages payment reconciliation and reviews income statements; reporting done monthly.
Mapping / Navigation Provider	Low	Low	Monitor	Integration that is easily replaceable; follow up on API status.
IoT Sensor Vendor	Low	Low	Monitor	Supplier of hardware pilot; manage the shipment and support.

Project Charter

Project Title	Smart Parking Platform
Project Sponsor	Consortium of municipal parking authorities & private garage operators (client)
Project Manager	Marcus Lee ($oftware Corp)
Prepared By	Imad - $oftware Corp
Start / End	September 2026 – April 2027
Purpose / Business Need	Driver searches for a park spot and waste time and fuel, and the operator lacks the real-time visibility and pricing power. Platform integrates the live availability, reservations, payments, and operator analytics.
Objectives	Develop driver's applications (web & mobile) and operator's dashboard including real-time availability, reservation, digital payment, reporting functionality meeting the SRS requirements.

High-Level Scope	In: application development, user interfaces, operator dashboard, data reporting. Out: hardware deployment, tickets generation, facility construction.
Key Deliverables	SRS; Work Breakdown Schedule and Schedule; product increments by Sprint; integrated platform; onboarding operators; reporting; handing-over documents.
Milestones	Requirements Baseline, Design complete, Authorization Module, Operator Dashboard, Integrations complete, Driver Application MVP, Beta Deployment, Full Deployment/hand-off .
Budget (BAC)	$2,517,120
Key Risks	Sensor unreliable data; payment gateway failure; integration and testing underestimate; key person loss
Key Stakeholders	Client sponsor, parking operators, drivers, transportation department of city, financial people, payment gateways, Software Corp teams.
Success Criteria	Acceptance of all deliverables conforming to SRS, timely and on budget delivery, operator's ability to on-board and driver's ability to pay and navigate.
Approval	The sponsor and Project Manager approval starts the project and authorizes Project Manager using the resources.

Team Charter

Team Mission	Ensure the construction and delivery of the Smart Parking Platform to an appropriate standard of quality and security, on time, as one collaborative team.
Team Scope	Cross-functional $oftware Corp Delivery team (18 people from product, development, design, QA, DevOps, security departments) is accountable for the design,

	creation, testing, and deployment of the Smart Parking Platform.
Goals	Provide each sprint with a work increment; maintain main in a state that can be released; achieve SRS objectives of quality and security; maintain a sustainable pace.
Roles & Responsibilities	Product Owner – priorities the backlog; Scrum Master facilitates the meetings; Architect designs the solution; Developers construct; QA tests; DevOps provides CI/CD; Security maintains the compliance (full list of RACI can be seen in §7.4).
Values	Transparency; respect; ownership; quality first; psychological safety - mistakes should be revealed at early stages and considered a learning opportunity.
Norms	Stand-up meetings; small reviewed Pull Requests; meaningful commits; enforced definition of done; no direct pushes to main branch; decision making documentation.
Decision-Making	Daily decisions about technical questions are taken by the relevant team in coordination with an Architect; backlog priorities are determined by the Product Owner; conflicts between teams are escalated to the PM. Default position – "disagree and commit."
Conflict Resolution	Deal with issues upfront; prefer cooperation and compromises; Scrum Master mediate; escalate unresolved conflicts to the PM.
Communication	Stand-up meetings, Sprint ceremonies, a joint Jira board, Github for coding and PR's, weekly risk review (see §6.5).
Metrics for Success	Sprint velocity and burndown chart, defect density, uptime, and on time/budget delivery (SPI/CPI).

Project Closure Report (Internal)

Executive Summary

The Smart Parking Platform project by $oftware Corp for the client consortium spanned eight months (September 2026 - April 2027) and achieved the goal of implementing a unified, real-time platform that allows the drivers to search, book, and pay for parking spaces and allows the operator to monitor the occupancy, set up dynamic prices and reporting features. The project completed one week later than scheduled, but only 2.6% over the budget; both parameters are within the range of tolerance, the project passed a security audit and penetration testing. Three low-priority reporting requirements were moved into a post-launch follow-up. As a result, the project can be defined as successful and recommended for client acceptance and closing.

Objectives & Deliverables

All the main objectives listed in the project charter were achieved: the driver's applications and websites, the operator's dashboard, the integrations with IoT devices, payment systems and mapping, the report, and the security and auditing components were implemented and accepted. All deliverables - SRS, WBS and schedule, sprint increments, the platform, the process of operator's onboarding and the documents for handover have been completed and documented.

Measure	Planned / Baseline	Actual	Variance
Schedule (finish)	Apr 17, 2027	Apr 24, 2027	+1 week (SPI ≈ 0.98)
Total cost	$2,517,120 (BAC)	$2,583,000	+2.6% (CPI ≈ 0.97)
Scope (backlog)	52 user stories	49 delivered, 3 deferred	Reporting extras moved to fast-follow
Quality (uptime)	99.9% target	99.92% in beta	Met
Security	PCI-DSS, pen-test	Passed audit & pen-test	Met

Successes and Failures

Successes

• Development of a consolidated and real-time driver and operator platform - that differentiator uncovered in competitive analysis.

• Full coverage of security requirements and compliance - including scope, encryption, penetration testing and the audit.
• Consolidation of 99.92% uptime in beta version versus the goal of 99.9%. Also - sub-2 second availability data updates from sensors.
• Delivery of 49 of 52 planned stories; the onboarding of facilities by operators was completed in the pilot successfully.
• Viable and sustainable pace without critical attrition of the team.

Failures
• Delivery was delayed 1 week and budget was overrun 2.6% due to underestimated integration/testing effort.
• Three stories for advanced reporting (price-performance analytics, peak hour calculations, and city data feed) were postponed to a fast follow release.
• On-street availability is available in real-time only for facilities having sensors. Otherwise the system uses a fall-back to the availability prediction model.
• Timely third-party access has been postponed, reducing the time for integration.

Lessons Learned
• Integrate testing explicitly in the budget. The largest deviation was because of underestimated integration/QA effort - the risk we flagged but under-resourced. Future plans should include additional schedule for this task.
• Request the third-party API access immediately. Postponed access delayed integration; request sandbox access on sprint 1 and mock the API calls before receiving credentials.
• Risk register paid off! Mitigations of the sensor feed risk (health checks and fallback to predictions), and of payment gateway (adapter and secondary provider), mitigated the two Extreme risks.
• Simultaneous operation of parallel teams and PR links ensured visibility in development process. PRs for backlog items kept the board clean and allowed to report achievements easily to the stakeholders.
• Clear and regular communication developed trust. Sprint review and weekly update reports helped avoid surprises at the closure point.

Recommendations For Future Projects
• Schedule explicitly the integration-and-hardening buffer (10-15% of overall time) for every project that relies on third-party systems.

• Make access to the third-party services (API keys, any hardware) an exit criterion of sprint 1 and use mocks otherwise.
• Keep and review periodically the risk register every second week and adopt its mitigations to the next project plan.
• Do not prefer an initial big chunk (full feature set), prefer a thin slice end to end (authorization + one facility + one reservation flow).
• Keep the lessons-learned repository updated to inherit experience from previous projects.

Acknowledgements And Sign-off

The project team deserves the appreciation for excellent delivery results. In order to finalize the project, approval from the Project Manager and the sponsor is required. It verifies that all deliverables were signed off and budget was reconciled.

Prepared by: Imad, $oftware Corp •
Approved by: _____ (Project Manager) _____ (Client Sponsor)